

IT1223(P) - Database Management Systems (P)
Department of Physical Science

University of Vavuniya

Lab Sheet - 03

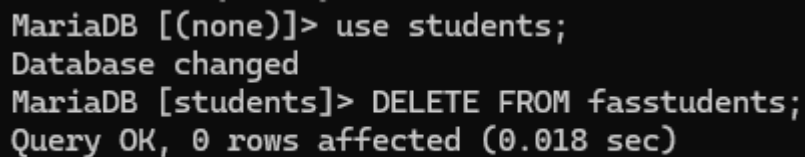
Answers

Q6) Continuation of Q5

1. How can you delete all records from the table fasstudents and how can you reload the same records again?

Query:

DELETE FROM fasstudents;



```
MariaDB [(none)]> use students;  
Database changed  
MariaDB [students]> DELETE FROM fasstudents;  
Query OK, 0 rows affected (0.018 sec)
```

```
INSERT INTO fasstudents (First_Name, Last_Name, Address, Sex, Date_of_Birth, Course,  
Telephone) VALUES  
  
('Nimal','Perera','10A, Galle Road, Colombo','M','1990-05-15','Computer Science','011- 2345678'),  
  
('Kamala','Silva','25, Kandy Road, Gampaha','F','1992-09-20','IT','033-4567890'),  
  
('Sunil','Fernando','30, Negombo Road, Negombo','M','1988-03-10','IT','031-3456789'),  
  
('Anula','Rajapakse','5B, Matara Street, Matara','F','1995-12-03','Computer Science','041-5678901'),  
  
('Malith','Jayawardena','15, Kurunegala Road, Kurunegala','M','1993-07-18','IT','037-6789012'),  
  
('Sujatha','Bandara','20, Jaffna Road, Jaffna','F','1991-11-25','Computer Science','021-7890123'),  
  
('Chamara','Rathnayake','35, Ratnapura Road, Ratnapura','M','1994-08-12','IT','045-8901234'),  
  
('Lakmini','Wijesinghe','8C, Badulla Street, Badulla','F','1989-06-30','Computer Science','055-  
9012345'),  
  
('Upul','Dissanayake','40, Ampara Road, Ampara','M','1996-02-14','Computer Science','063-  
0123456'),  
  
('Tharanga','Perumal','55, Polonnaruwa Road, Polonnaruwa','F','1997-04-05','IT','027-1234567');
```

```

MariaDB [students]> CREATE TABLE fasstudents(
  -> First_Name VARCHAR(25),
  -> Last_Name VARCHAR(25),
  -> Address VARCHAR(40),
  -> Sex CHAR(1),
  -> Date_of_Birth DATE,
  -> Course VARCHAR(25),
  -> Telephone VARCHAR(10)
  -> );
Query OK, 0 rows affected (0.024 sec)

MariaDB [students]> INSERT INTO fasstudents (First_Name, Last_Name, Address, Sex, Date_of_Birth, Course, Telephone) VALUES
  -> ('Nimal','Perera','10A, Galle Road, Colombo','M','1990-05-15','Computer Science','011- 2345678'),
  -> ('Kamala','Silva','25, Kandy Road, Gampaha','F','1992-09-20','IT','033-4567890'),
  -> ('Sunil','Fernando','30, Negombo Road, Negombo','M','1988-03-10','IT','031-3456789'),
  -> ('Anula','Rajapakse','5B, Matara Street, Matara','F','1995-12-03','Computer Science','041-5678901'),
  -> ('Malith','Jayawardena','15, Kurunegala Road, Kurunegala','M','1993-07-18','IT','037-6789012'),
  -> ('Sujatha','Bandara','20, Jaffna Road, Jaffna','F','1991-11-25','Computer Science','021-7890123'),
  -> ('Chamara','Rathnayake','35, Ratnapura Road, Ratnapura','M','1994-08-12','IT','045-8901234'),
  -> ('Lakmini','Wijesinghe','8C, Badulla Street, Badulla','F','1989-06-30','Computer Science','055-9012345'),
  -> ('Upul','Dissanayake','40, Ampara Road, Ampara','M','1996-02-14','Computer Science','063-0123456'),
  -> ('Tharanga','Perumal','55, Polonnaruwa Road, Polonnaruwa','F','1997-04-05','IT','027-1234567');
Query OK, 10 rows affected, 10 warnings (0.010 sec)
Records: 10 Duplicates: 0 Warnings: 10

```

2. Display all students first name, date of birth and their current ages.

Query:

```

SELECT First_Name, Date_of_Birth, YEAR(CURDATE())-YEAR (Date_of_Birth) AS Age FROM
fasstudents;

```

```

MariaDB [students]> SELECT First_Name, Date_of_Birth, YEAR(CURDATE())-YEAR (Date_of_Birth) AS Age FROM fasstudents;
+-----+-----+-----+
| First_Name | Date_of_Birth | Age |
+-----+-----+-----+
| Nimal      | 1990-05-15    | 34  |
| Kamala     | 1992-09-20    | 32  |
| Sunil      | 1988-03-10    | 36  |
| Anula      | 1995-12-03    | 29  |
| Malith     | 1993-07-18    | 31  |
| Sujatha    | 1991-11-25    | 33  |
| Chamara    | 1994-08-12    | 30  |
| Lakmini    | 1989-06-30    | 35  |
| Upul       | 1996-02-14    | 28  |
| Tharanga   | 1997-04-05    | 27  |
+-----+-----+-----+
10 rows in set (0.001 sec)

```

3. How can you sort the output in (ii) by first name?

Query:

```

SELECT First_Name, Date_of_Birth, YEAR(CURDATE())-YEAR (Date_of_Birth) AS Age FROM fasstudents
ORDER BY First_Name;

```

```
MariaDB [students]> SELECT First_Name, Date_of_Birth, YEAR(CURDATE())-YEAR (Date_of_Birth) AS Age FROM fasstudents
ORDER BY First_Name;
```

First_Name	Date_of_Birth	Age
Anula	1995-12-03	29
Chamara	1994-08-12	30
Kamala	1992-09-20	32
Lakmini	1989-06-30	35
Malith	1993-07-18	31
Nimal	1990-05-15	34
Sujatha	1991-11-25	33
Sunil	1988-03-10	36
Tharanga	1997-04-05	27
Upul	1996-02-14	28

10 rows in set (0.001 sec)

4. How can you sort the output in (i) by age?

Query:

```
SELECT First_Name, Date_of_Birth, YEAR(CURDATE())-YEAR (Date_of_Birth) AS Age FROM fasstudents
ORDER BY Age;
```

```
MariaDB [students]> SELECT First_Name, Date_of_Birth, YEAR(CURDATE())-YEAR (Date_of_Birth) AS Age FROM fasstudents ORDER BY Age;
BY Age;
```

First_Name	Date_of_Birth	Age
Tharanga	1997-04-05	27
Upul	1996-02-14	28
Anula	1995-12-03	29
Chamara	1994-08-12	30
Malith	1993-07-18	31
Kamala	1992-09-20	32
Sujatha	1991-11-25	33
Nimal	1990-05-15	34
Lakmini	1989-06-30	35
Sunil	1988-03-10	36

10 rows in set (0.001 sec)

5. How can you display all students first name and their birth month?

Query:

```
SELECT First_Name, MONTH(Date_of_Birth) AS Birth_Month FROM fasstudents;
```

```
MariaDB [students]> SELECT First_Name, MONTH(Date_of_Birth) AS Birth_Month FROM fasstudents;
```

First_Name	Birth_Month
Nimal	5
Kamala	9
Sunil	3
Anula	12
Malith	7
Sujatha	11
Chamara	8
Lakmini	6
Upul	2
Tharanga	4

10 rows in set (0.001 sec)

6. How can you display all students who were born in month April?

Query:

```
SELECT * FROM fasstudents WHERE MONTH (Date_of_Birth) = 4;
```

```
MariaDB [students]> SELECT * FROM fasstudents WHERE MONTH (Date_of_Birth) = 4;
```

First_Name	Last_Name	Address	Sex	Date_of_Birth	Course	Telephone
Tharanga	Perumal	55, Polonnaruwa Road, Polonnaruwa	F	1997-04-05	IT	027-123456

```
1 row in set (0.001 sec)
```

SQL pattern matching enables you to use “_” to match any single character and “%” to match an arbitrary number of characters (including zero characters).

7. Display all students names beginning with “M”.

Query:

```
SELECT * FROM fasstudents WHERE First_Name LIKE 'M%';
```

```
MariaDB [students]> SELECT * FROM fasstudents WHERE First_Name LIKE 'M%';
```

First_Name	Last_Name	Address	Sex	Date_of_Birth	Course	Telephone
Malith	Jayawardena	15, Kurunegala Road, Kurunegala	M	1993-07-18	IT	037-678901

```
1 row in set (0.001 sec)
```

8. Display all students names ending with “NA”.

Query:

```
SELECT * FROM fasstudents WHERE Last_Name LIKE '%NA';
```

```
MariaDB [students]> SELECT * FROM fasstudents WHERE Last_Name LIKE '%NA';
```

First_Name	Last_Name	Address	Sex	Date_of_Birth	Course	Telephone
Malith	Jayawardena	15, Kurunegala Road, Kurunegala	M	1993-07-18	IT	037-678901

```
1 row in set (0.001 sec)
```

9. Display all students names containing exactly six characters.

Query:

```
SELECT * FROM fasstudents WHERE LENGTH (First_Name)=6;
```

```
MariaDB [students]> SELECT * FROM fasstudents WHERE LENGTH (First_Name)=6;
```

First_Name	Last_Name	Address	Sex	Date_of_Birth	Course	Telephone
Kamala	Silva	25, Kandy Road, Gampaha	F	1992-09-20	IT	033-456789
Malith	Jayawardena	15, Kurunegala Road, Kurunegala	M	1993-07-18	IT	037-678901

```
2 rows in set (0.001 sec)
```

10. Repeat parts vii,viii (above) using Regular expressions.

Query:

```
SELECT * FROM fasstudents WHERE First_Name REGEXP '^M';
```

```
MariaDB [students]> SELECT * FROM fasstudents WHERE First_Name REGEXP '^M';
```

First_Name	Last_Name	Address	Sex	Date_of_Birth	Course	Telephone
Malith	Jayawardena	15, Kurunegala Road, Kurunegala	M	1993-07-18	IT	037-678901

```
1 row in set (0.001 sec)
```

SELECT * FROM fasstudents WHERE Last_Name REGEXP 'NA\$';

```
MariaDB [students]> SELECT * FROM fasstudents WHERE Last_Name REGEXP 'NA$';
+-----+-----+-----+-----+-----+-----+-----+
| First_Name | Last_Name | Address | Sex | Date_of_Birth | Course | Telephone |
+-----+-----+-----+-----+-----+-----+-----+
| Malith | Jayawardena | 15, Kurunegala Road, Kurunegala | M | 1993-07-18 | IT | 037-678901 |
+-----+-----+-----+-----+-----+-----+-----+
1 row in set (0.001 sec)
```

11. How can you find records where the firstnames containing the letter “o”.

Query:

SELECT * FROM fasstudents WHERE First_Name LIKE '%o%';

```
MariaDB [students]> SELECT * FROM fasstudents WHERE First_Name LIKE '%o%';
Empty set (0.001 sec)
```

```
MariaDB [students]> SELECT COUNT(*) AS Student_Count FROM fasstudents;
```

12. How can you count the number of students in the fasstudents table.

Query:

SELECT COUNT(*) AS Student_Count FROM fasstudents;

```
MariaDB [students]> SELECT COUNT(*) AS Student_Count FROM fasstudents;
+-----+
| Student_Count |
+-----+
| 10 |
+-----+
1 row in set (0.000 sec)
```